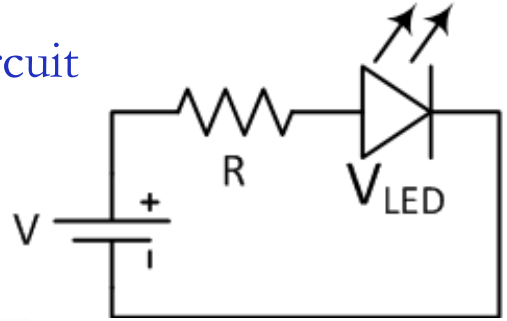
Light Emitting Diodes

The following subjects will be covered

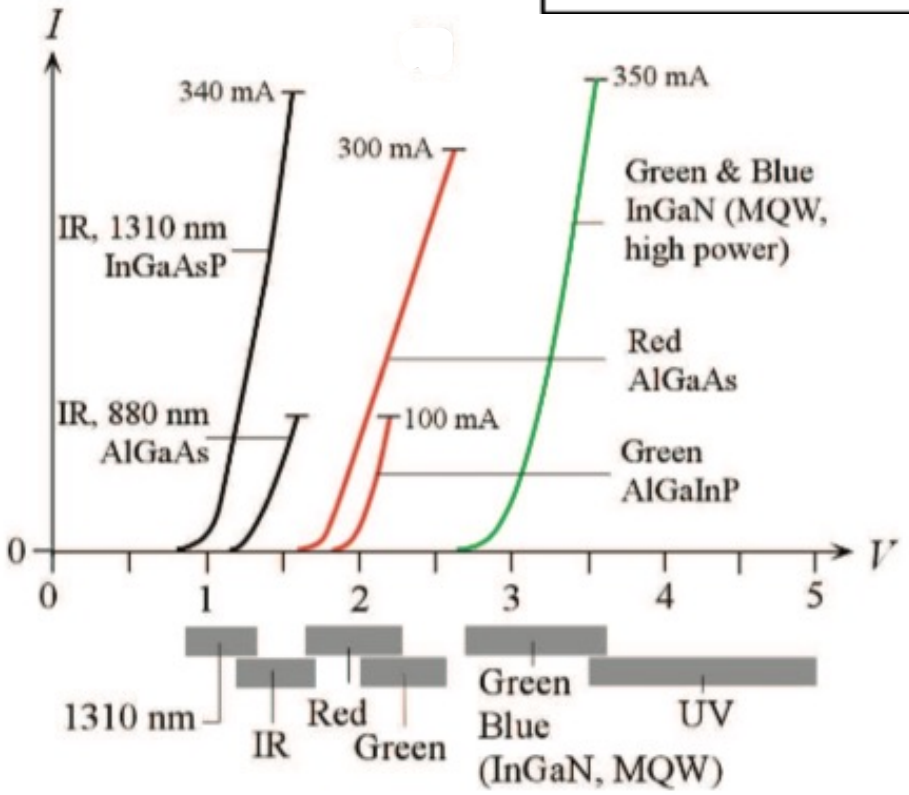
- The PN Junction Band Diagram.
- LED Basic Design and Operation.
- LED Efficiency.
- Improved LED Designs.
- Basic LED Types.
- LED Characterization.
- LED Materials.

LED Characterization: Output Optical Power

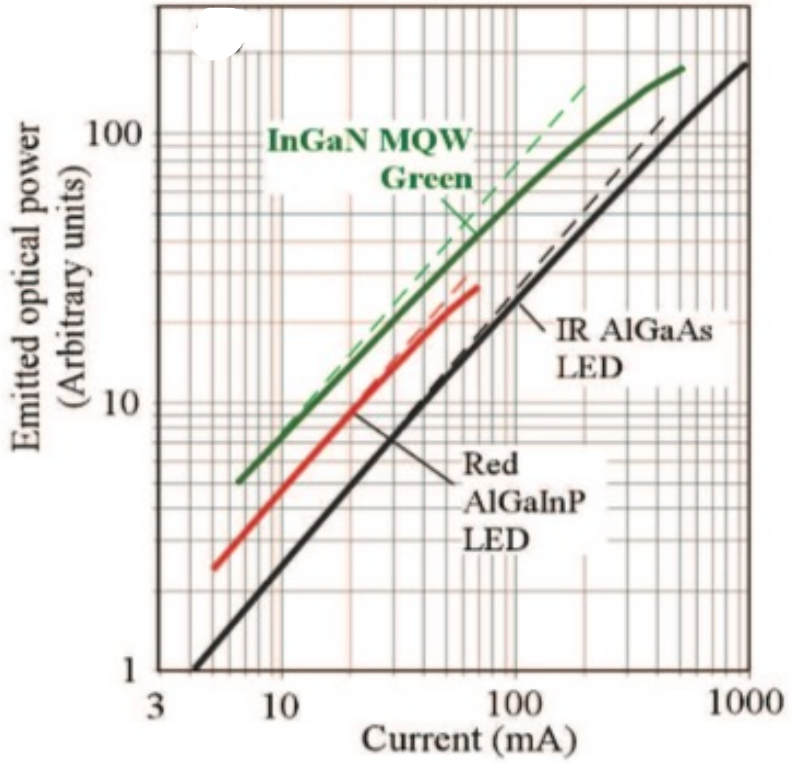
Circuit



IV Curve



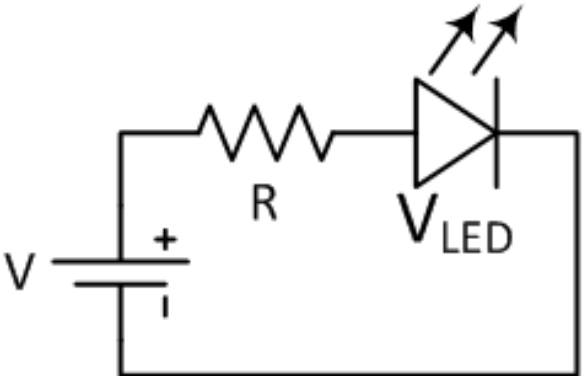
PI Curve $P_o = \hbar\omega\, \eta_{ex}\, \frac{i}{q}$

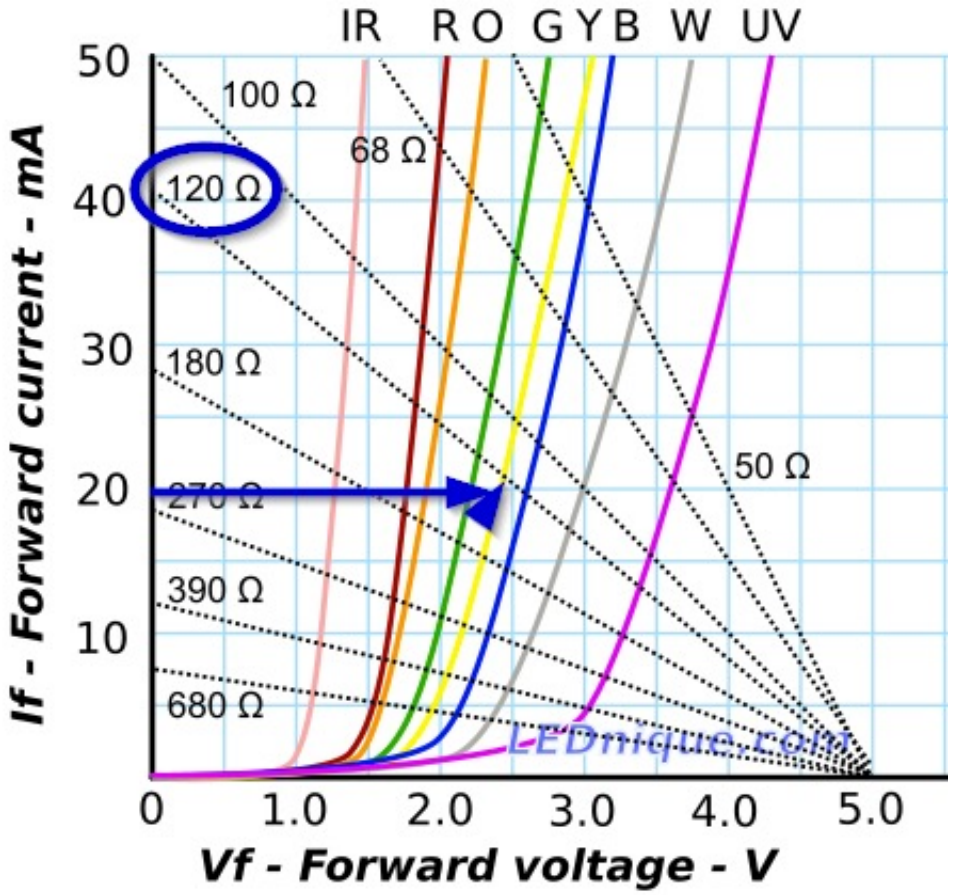


- Carrier leakage
- Joule heating

LED Characterization: Output Optical Power

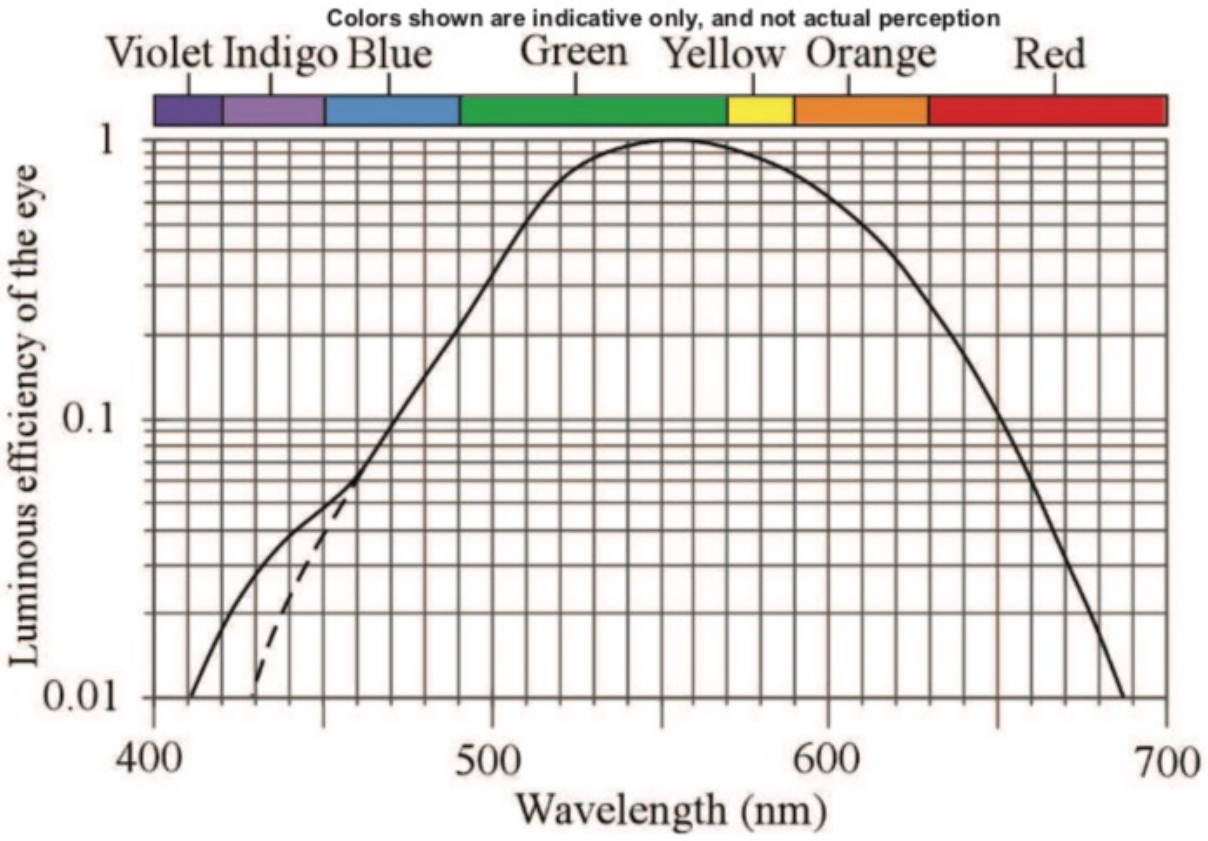
Example: Find the value of the series resistance for a yellow LED biased at 20 mA. Take V=5 V.





LED Characterization: Output Optical Power

The human eye optical spectrum: Luminous efficiency.



Example: What is the luminous flux of a red LED emitting at 650 nm with output power of 10 mW?

Luminous efficiency at 650 nm is 0.1.

Luminous flux is given by

$$Lum = \eta_{LE} P_o (633\ lm/W)$$

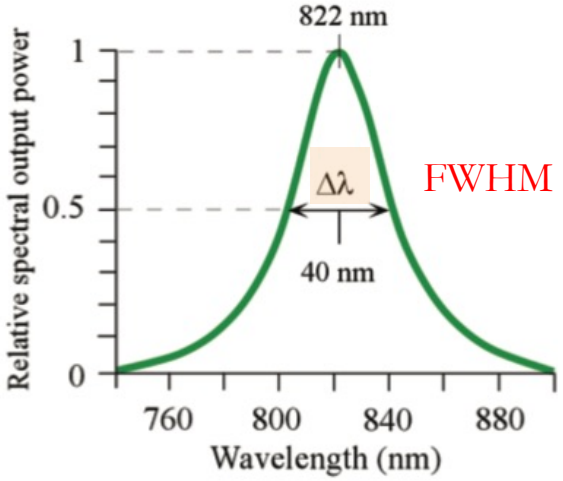
$$Lum = 0.63\ \text{lm}$$

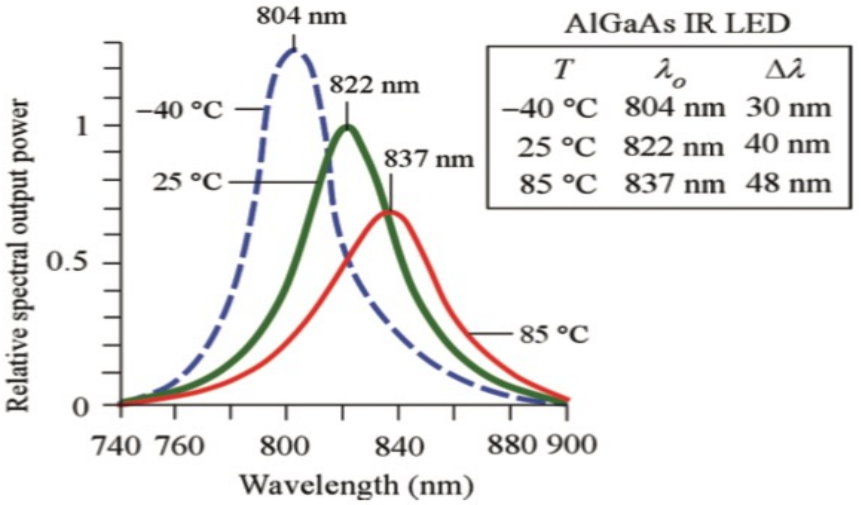
Repeat for green at 532 nm.

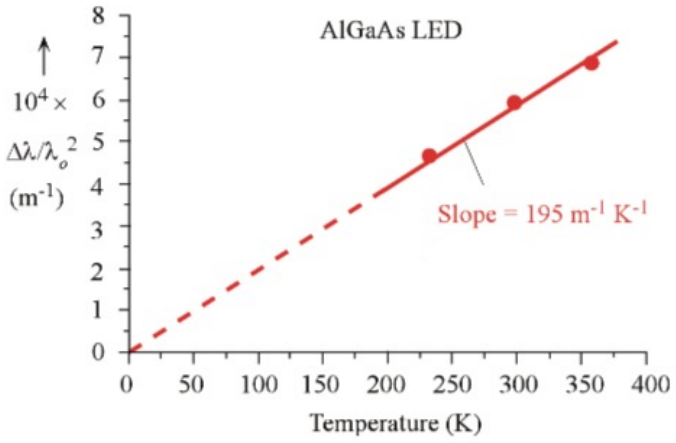
➡ $Lum = 5.51\ \text{lm}$

LED Characterization: Spectral Distribution

Spectral Distribution of an LED is described by peak wavelength λ_o and spectral width $\Delta\lambda$.







$$E_G = E_{Go} - \frac{AT^2}{(B+T)}$$ Varshni Equation

- Red-shift with temperature is due to decrease in bandgap.
- Increase of spectral width with temperature is due to electron-hole distribution in the conduction/valence bands.

LED Characterization: Spectral Distribution

We can express the spectral width $\Delta\lambda$ in term of change in photon energy $\Delta\hbar\omega$.

$$\lambda = \frac{hc}{E}$$

$$\frac{\Delta\lambda}{\Delta E} = hc\left(\frac{-1}{E^2}\right)$$

$$\Delta\lambda = -hc\left(\frac{\Delta E}{E^2}\right)$$

$$\Delta\lambda = -\left(\frac{\lambda^2}{hc}\right)\Delta E$$

$$\frac{\Delta\lambda}{\lambda^2} = -\frac{\Delta E}{hc}$$

$$\Delta E \approx 2.5 - 3\,k_B T$$

Example: Find the spectral width for a GaAsP LED emitting at 655 nm at 300 K. Take $\Delta E = 3\,k_B T$.

$$\Delta\lambda = \left(\frac{(655\times10^{-9})^2}{6.626\times10^{-34}\times3\times10^{8}}\right)(3\times1.38\times10^{-23}\times300)$$

$$\Delta\lambda = 26.8\,nm$$

Example: Find the shift in the emitted wavelength of a GaAs LED per 1 °C change at room temperature. E_{Go}=1.519 eV, A= 5.41×10^{-4} eV/K, B=204 K.

$$\frac{\Delta E_G}{\Delta T} = -\left(\frac{E_{Go} - E_G}{T}\right)\left(\frac{2B}{T+B}\right)$$

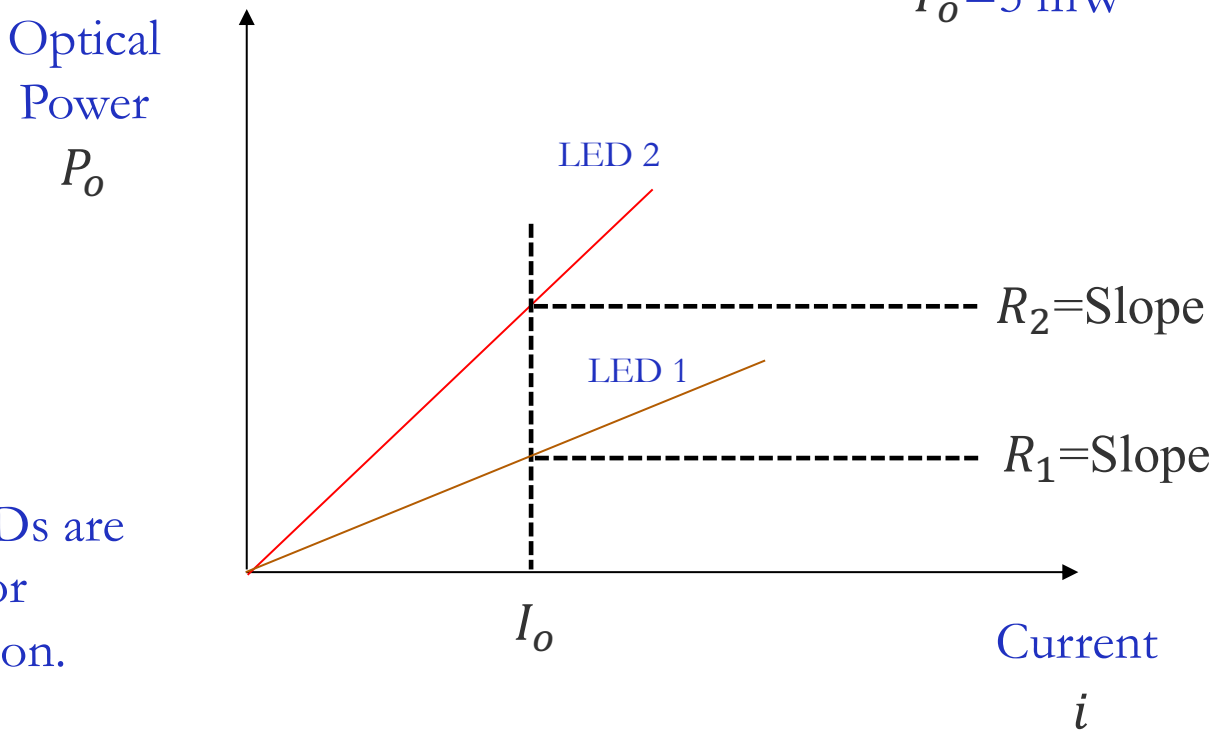
LED Characterization: Responsivity

LED Responsivity is defined as,

$$R = \frac{P_o}{i} = \frac{\hbar\omega}{q}\,\eta_{ex}$$

Example: An LED with R=50 μW/mA, find the output power for I_o=100 mA.

P_o=5 mW



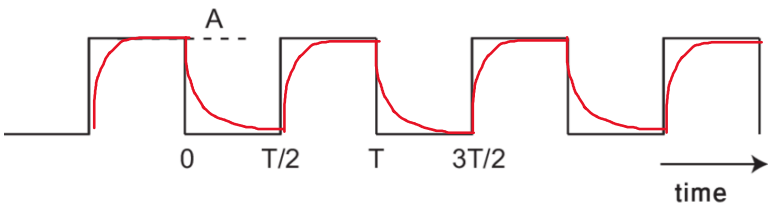
High R LEDs are used for modulation.

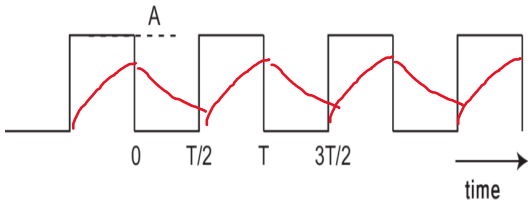
LED Characterization: Modulation Bandwidth

The ability of the LED to follow the speed of transitions of the AC signal.

Main factors

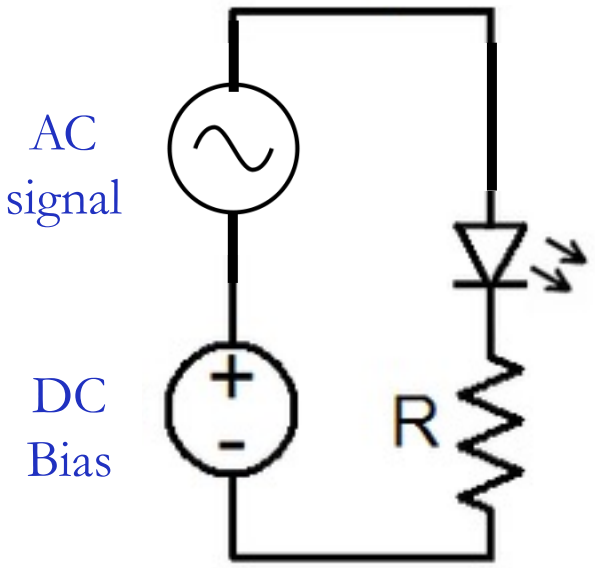
1) Carrier recombination time (optical).
2) Circuit RC delay (electrical).





Good communication LEDs have 300-700 MHz bandwidth

$\rightarrow i = I_{DC} + i_{ac}$

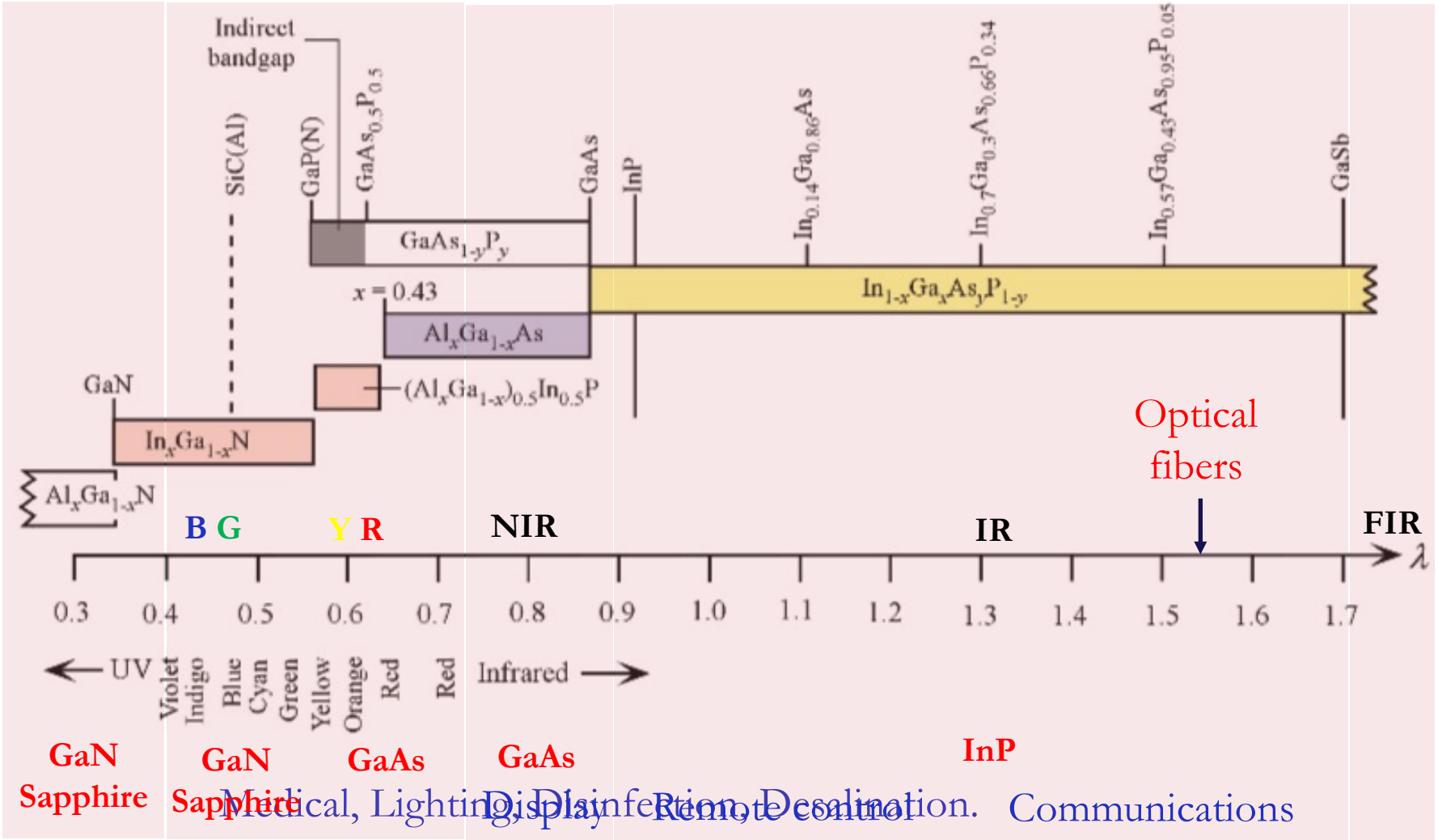


Optical cut-off frequency $f_c = \frac{\sqrt{3}}{2\pi\tau}$

Electrical cut-off frequency $f_c = \frac{1}{2\pi RC}$

LED Materials

Materials and Applications.

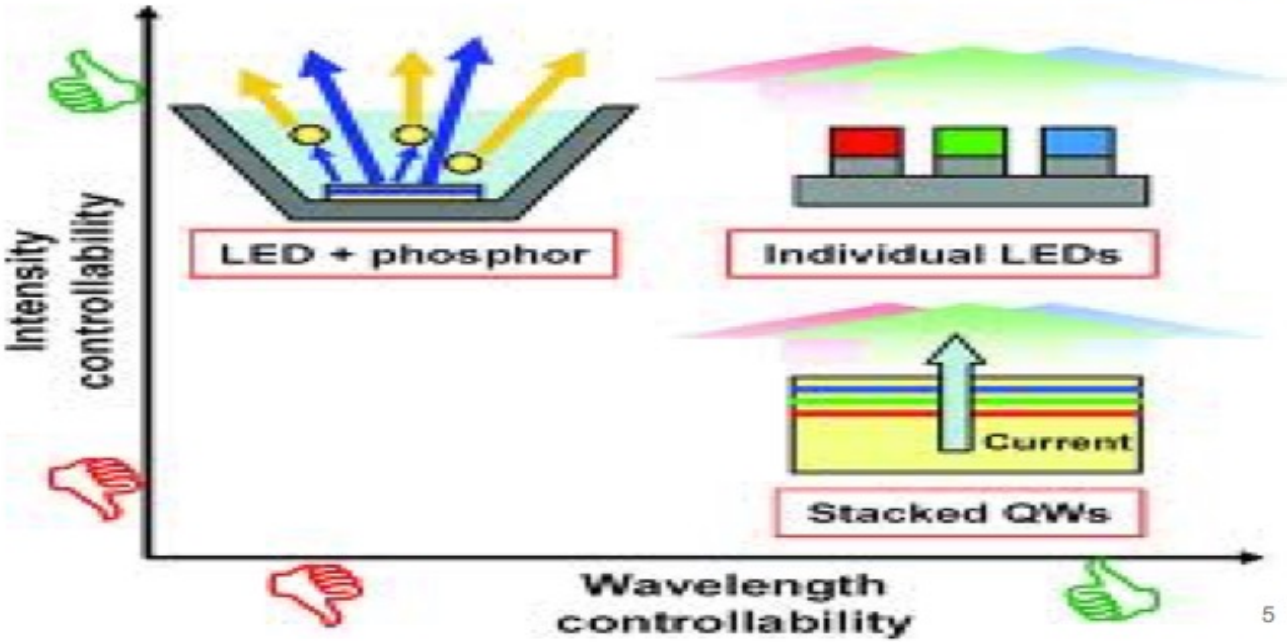


White LEDs

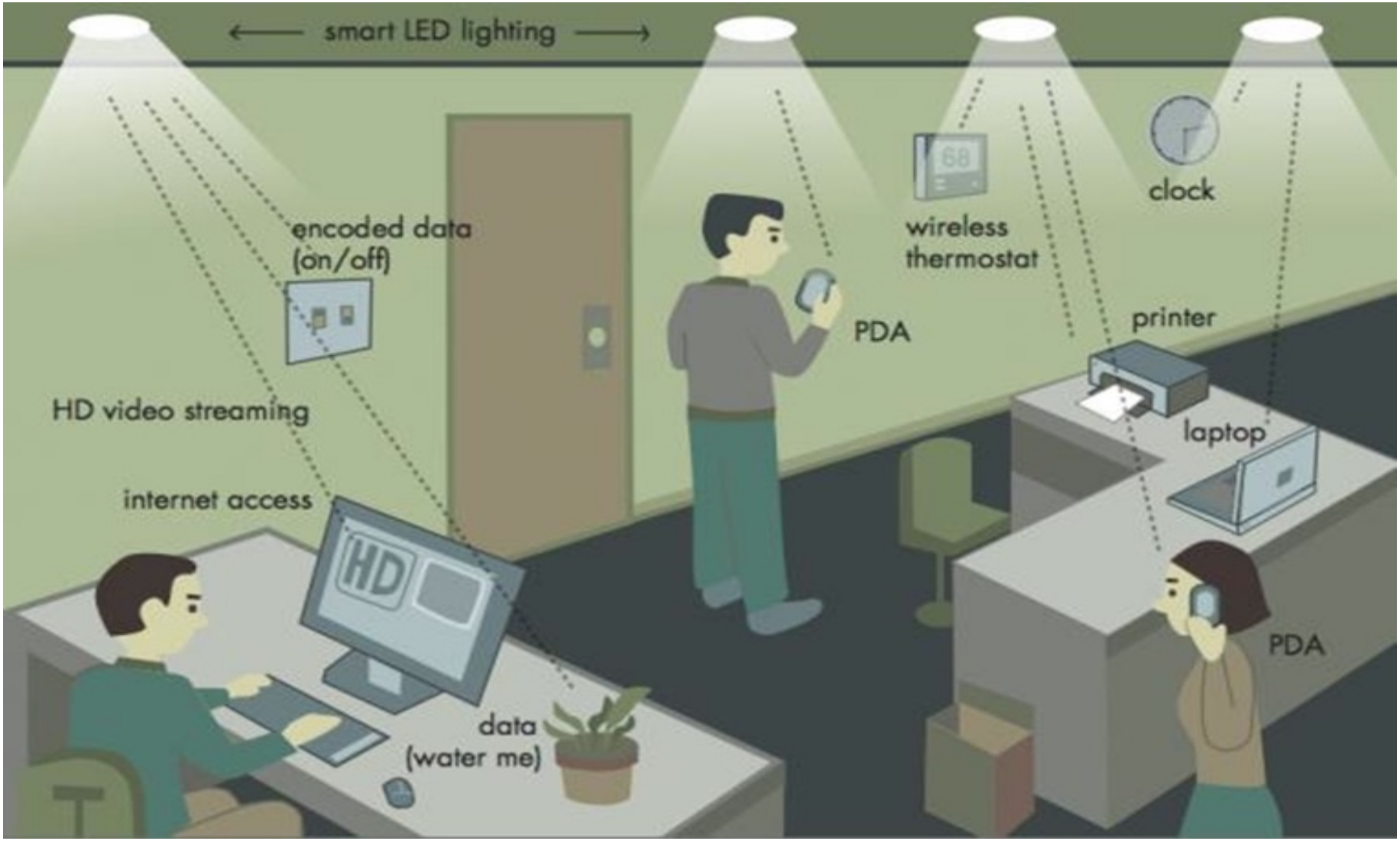
Two basic approaches:

1) Phosphor-based. High-energy LED is used as an excitation.
2) Multi-chip-based. Three individual LEDs are used (R-G-B).

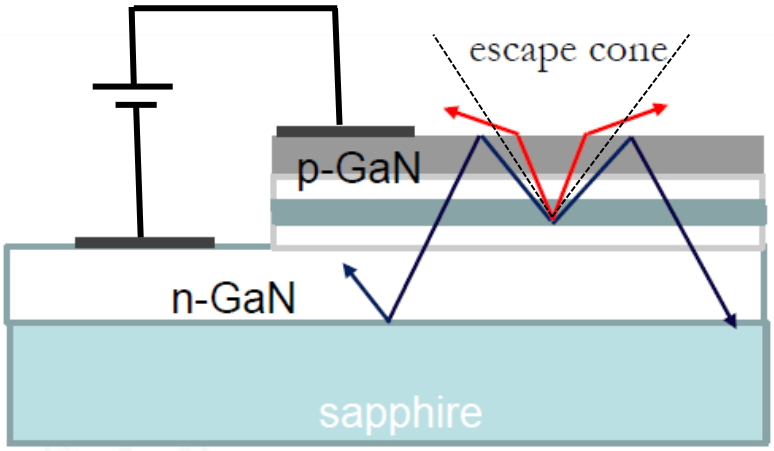




Solid-state Lighting



Solid-state Lighting

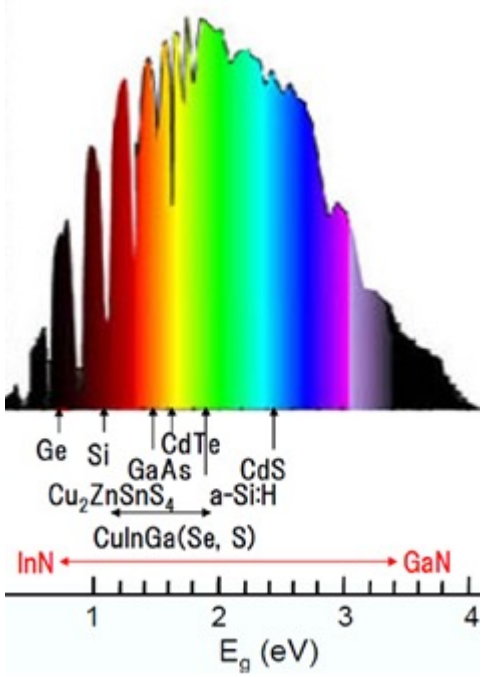


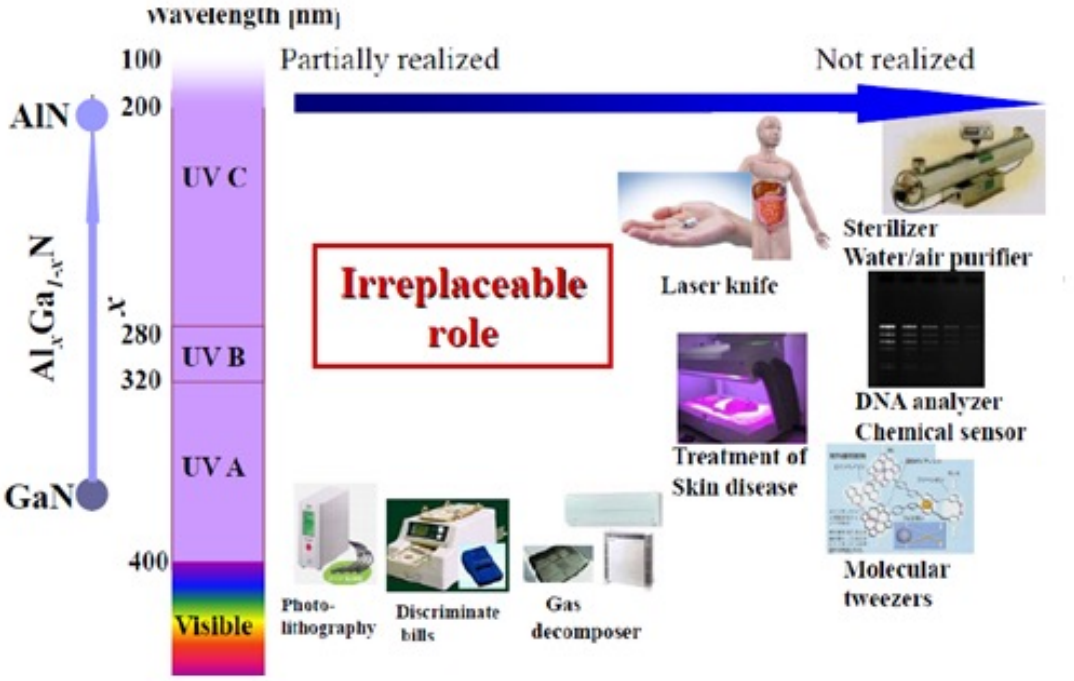
For energy-efficient white light solution, we need:

- ✓ Efficient photon generation: Internal Quantum Efficiency (IQE).
- ✓ Efficient photon extraction: External Quantum Efficiency (EQE).

Material as well as design challenges!

Solid-state Lighting

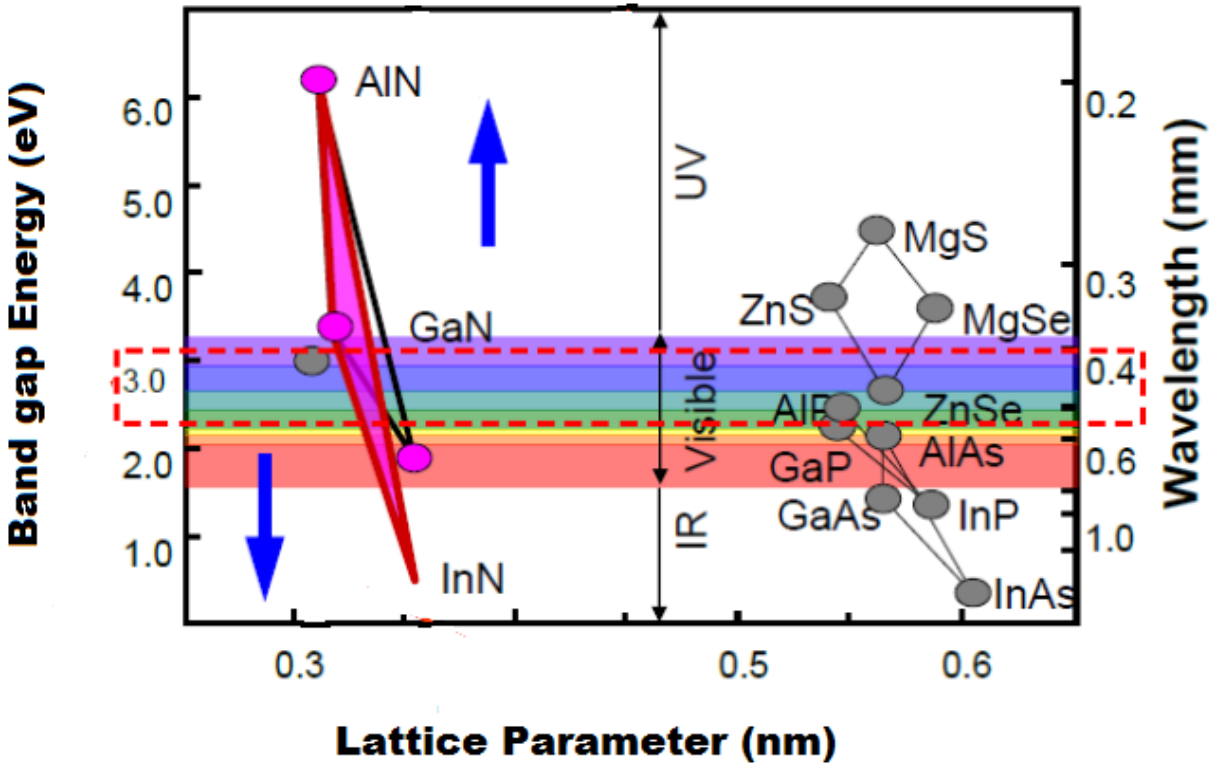




							VIIIA
		IIIA	IVA	VA	VIA	VIIA	2 He 4.003
		5 B 10.811	6 C 12.011	7 N 14.007	8 O 15.999	9 F 18.998	10 Ne 20.183
IB	IIB	13 Al 26.982	14 Si 28.086	15 P 30.974	16 S 32.064	17 Cl 35.453	18 Ar 39.948
29 Cu 63.54	30 Zn 65.37	31 Ga 69.72	32 Ge 72.59	33 As 74.922	34 Se 78.96	35 Br 79.909	36 Kr 83.80
47 Ag 107.870	48 Cd 112.40	49 In 114.82	50 Sn 118.69	51 Sb 121.75	52 Te 127.60	53 I 126.904	54 Xe 131.30
79 Au 196.967	80 Hg 200.59	81 Tl 204.37	82 Pb 207.19	83 Bi 208.980	84 Po (210)	85 At (210)	86 Rn (222)

- ✓ InGaN is direct-gap over 0.7-3 eV with high carrier mobility, allowing the realization of efficient solar cells even with high dislocation densities and high piezoelectric constant.
- ✓ Components made from other semiconductors can be replaced.

Solid-state Lighting



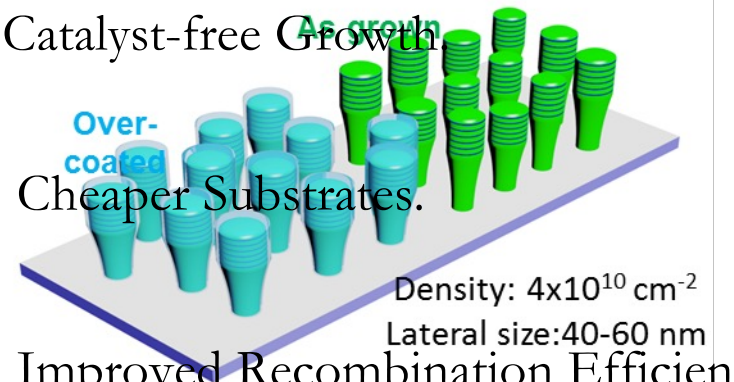
- ✓ Only a narrow III-N energy range is utilized.
- ✓ If more Al and In is incorporated, III-N can replace all other material systems.

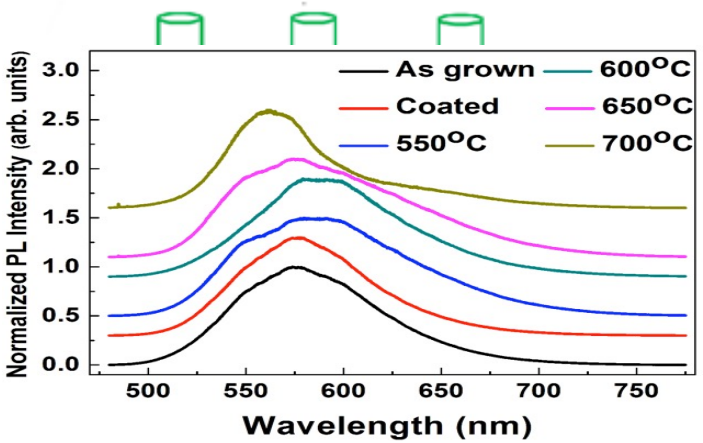
Planar Structure Design

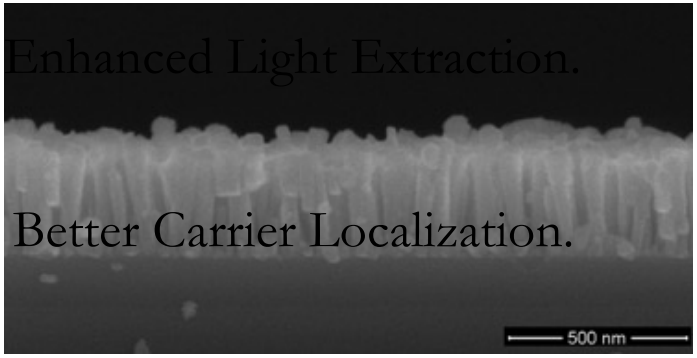
- Low Crystalline Quality.
  - ✓ Strong mismatch field effects.
  - ✓ Phase separation.
- Low Recombination Efficiency.
  - ✓ Strong polarization fields.
  - ✓ Optical anisotropy.
- Efficiency Droop.
  - ✓ Auger non-radiative recombination.
- Poor Light Extraction.
  - ✓ Total internal reflection.
  - ✓ Waveguiding effects.
- Carrier De-localization.
  - ✓ QW thickness.

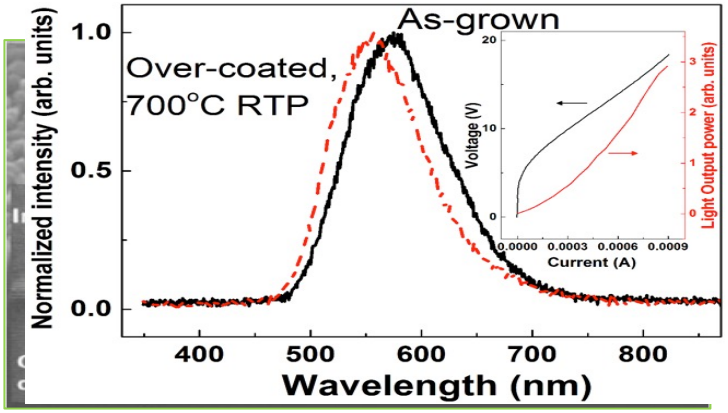
Quantum Disk in Nanowire

- Catalyst-free Growth.
- Cheaper Substrates.
- Improved Recombination Efficiency.
- Enhanced Light Extraction.
- Better Carrier Localization.

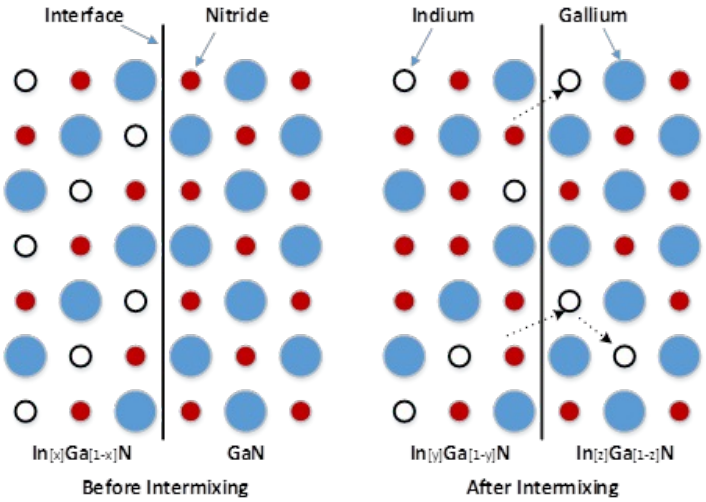


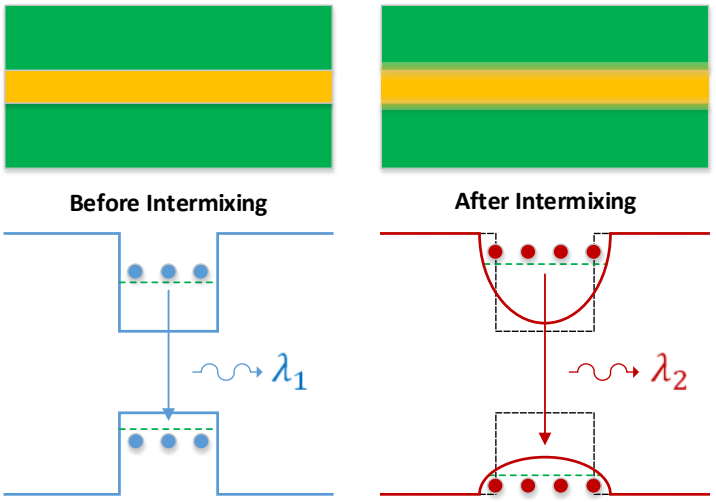






Intermixing





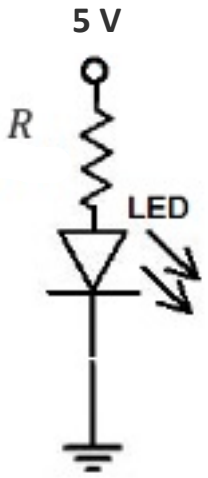
- ✓ Intermixing phenomenon at InGaN/GaN interface.
- ✓ Tune of the emission wavelength is possible. It is related to band gap energy.

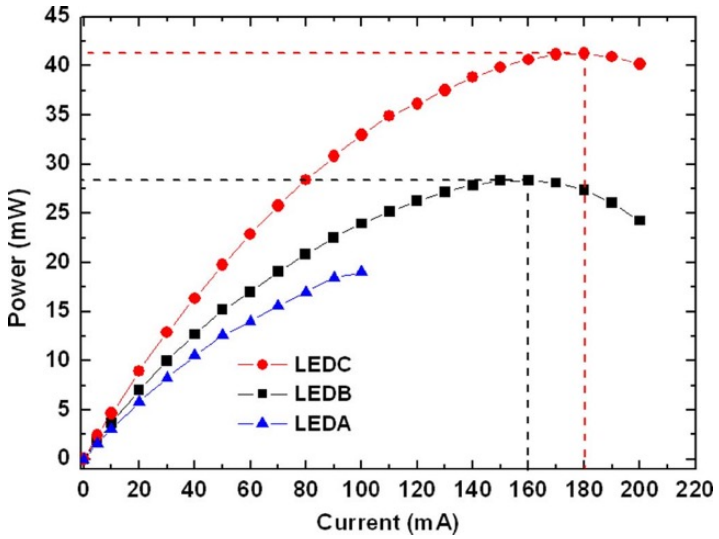
LED Circuits

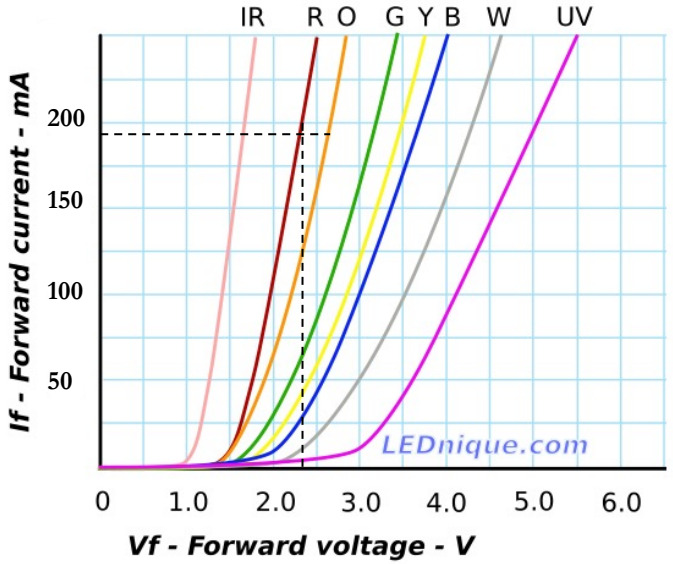
Biasing circuit for LED. The amount of current should

1) Be enough to switch On the LED
2) Correspond to the required brightness.

Find the value of R for maximum optical intensity from LEDC.







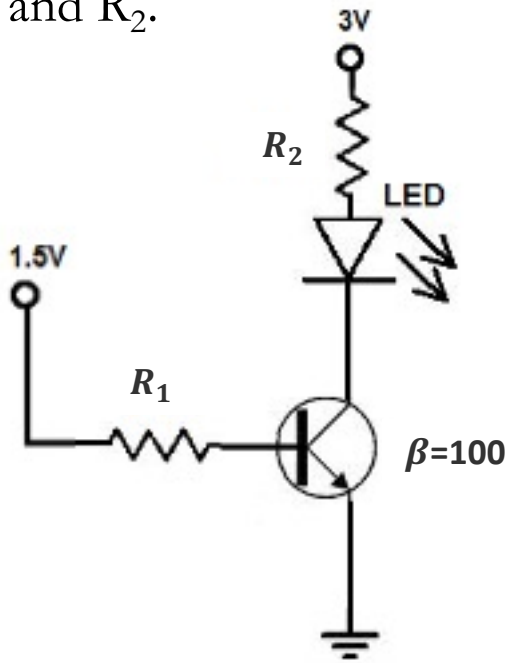
$$I = 180\ mA \qquad R = \frac{5 - 2.4}{180\ m} = 14\ \Omega$$

LED Circuits

Biasing circuit for LED. The amount of current should

1) Be enough to switch On the LED
2) Correspond to the required brightness.

Find the value of R_1 and R_2.



$$I_D = 180\ mA = I_C$$

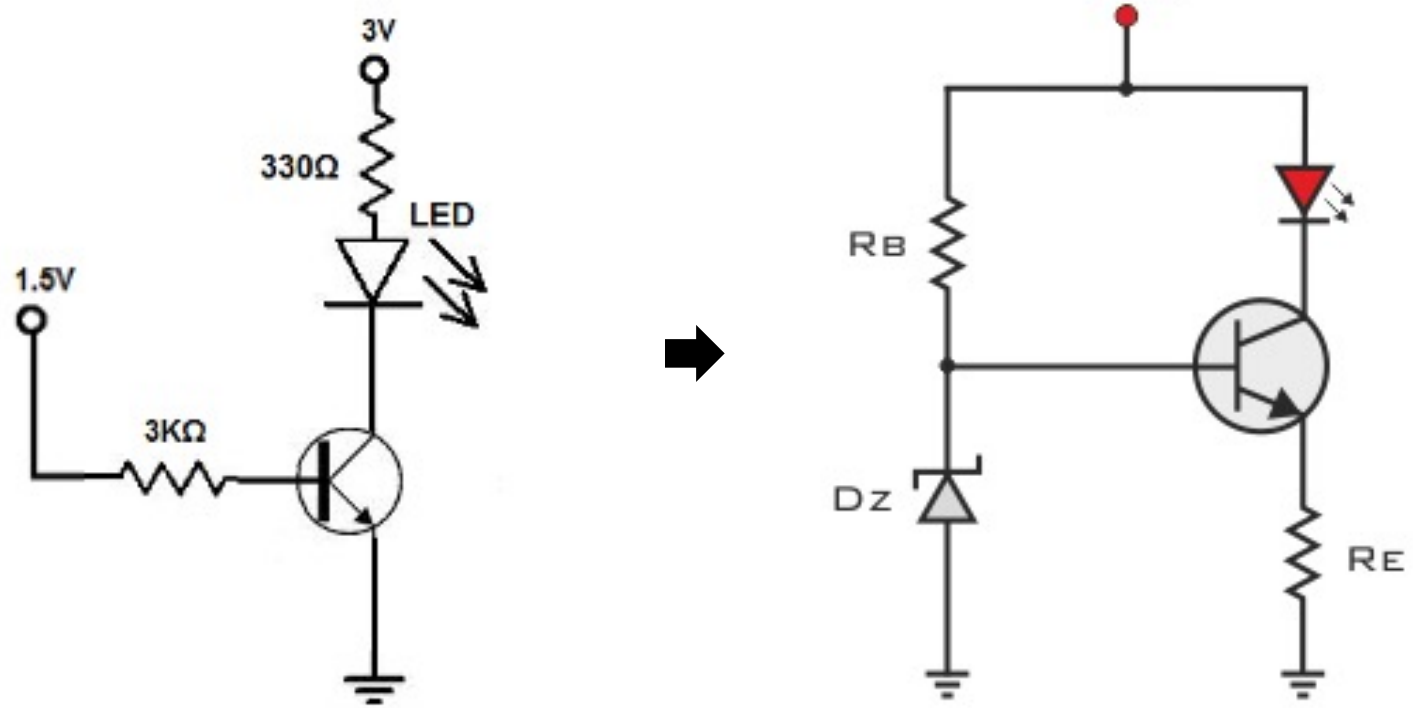
$$I_B = \frac{180\ m}{100} = 1.8\ mA$$

$$R_1 = \frac{1.5 - 0.7}{1.8\ m} = 440\ \Omega$$

$$R_2 = Any\ suitable\ value$$

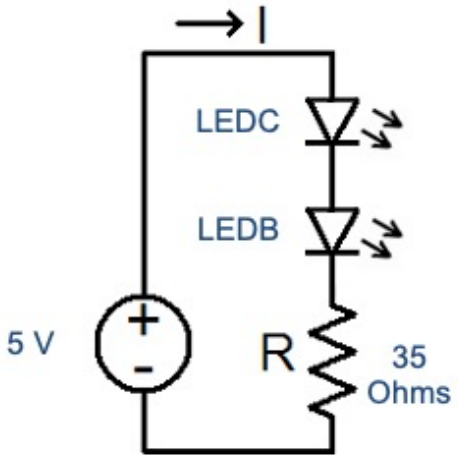
LED Circuits

LED light regulation circuits are used to keep light intensity constant against variations in input voltage.



LED Circuits

For the given LED circuit, find the luminous flux of each LED. Both diodes have a built-in voltage of 1 V. The PI curves of the diodes are as shown.



$$I = \frac{5-2}{35} = 85.7\ mA$$

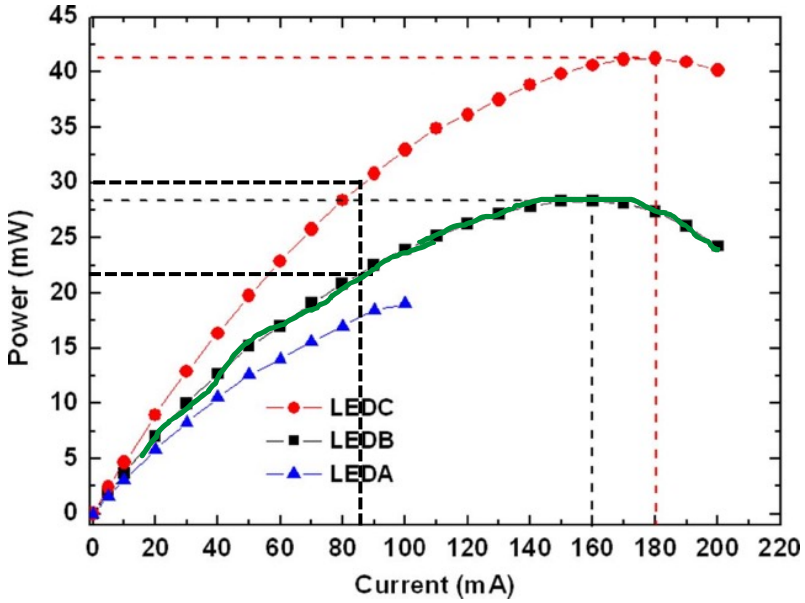
$P_{o_B} = 22\ mW$

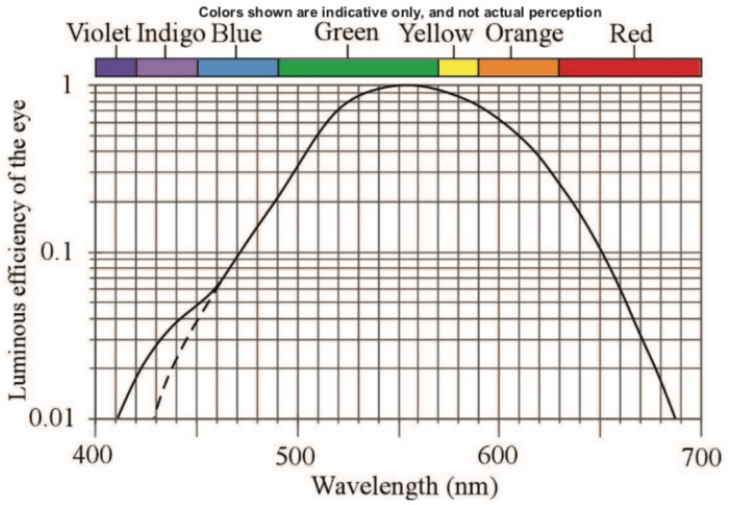
$P_{o_C} = 30\ mW$

$$Lum = \eta_{LE} P_o (633\ lm/W)$$

$$Lum_B = (1)(22\ m)\left(633\frac{lm}{W}\right) = 13.9$$

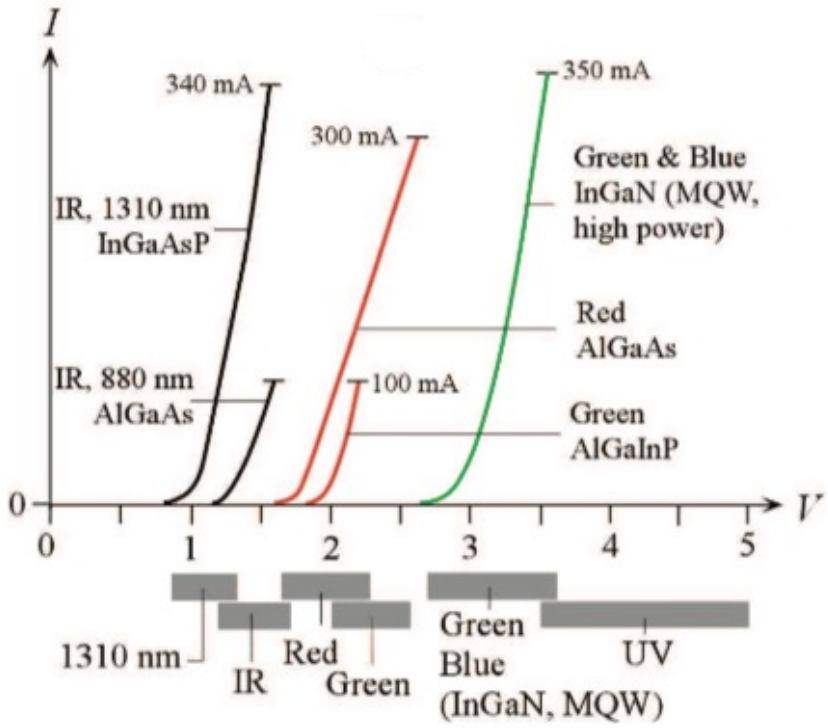
$$Lum_C = (0.1)(30\ m)\left(633\frac{lm}{W}\right) = 1.9$$

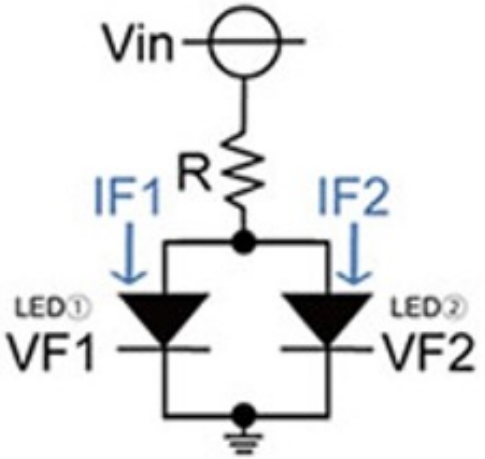


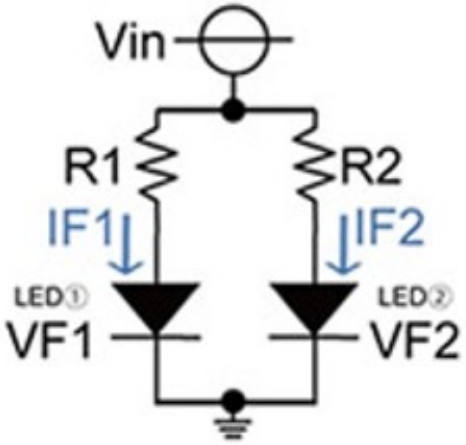


LED Circuits

Multiple LEDs circuit connection.

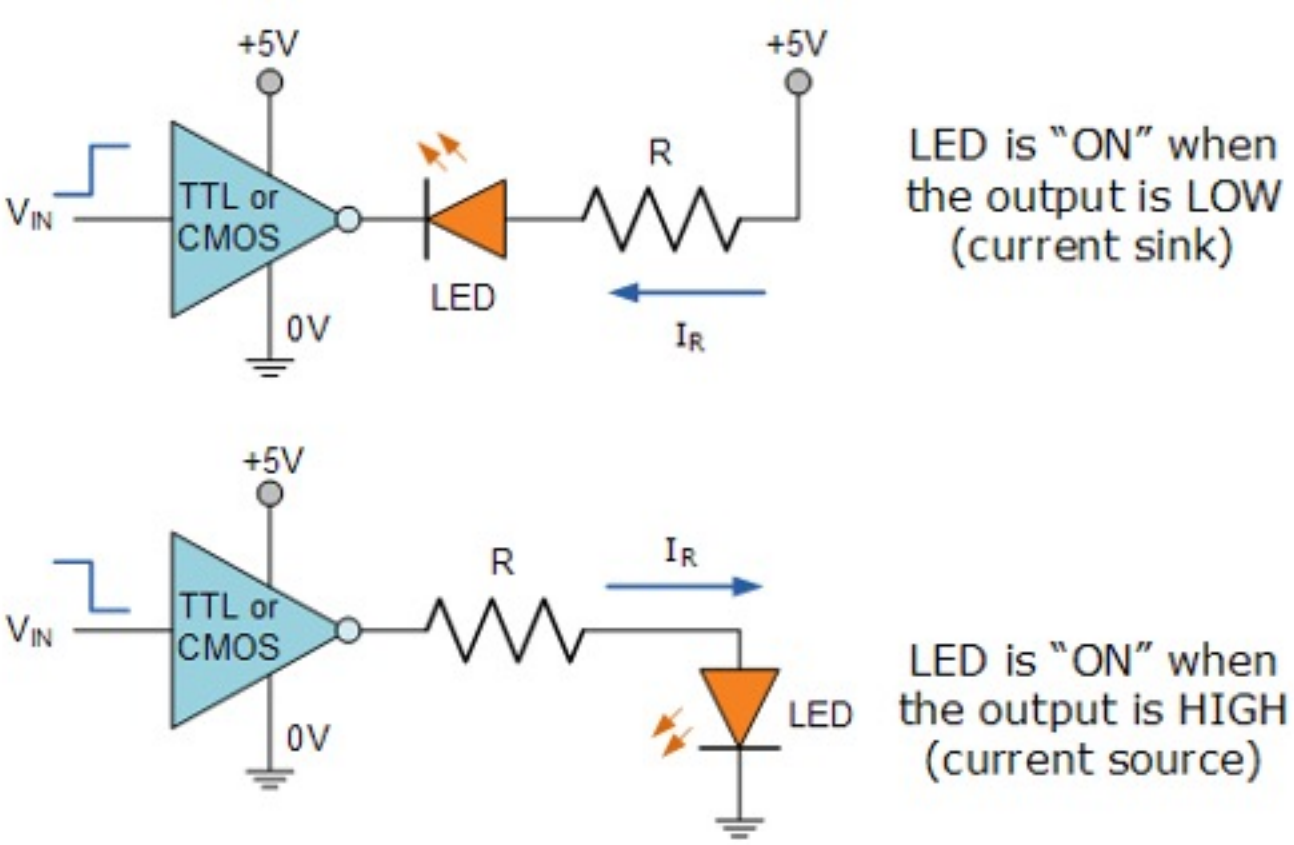






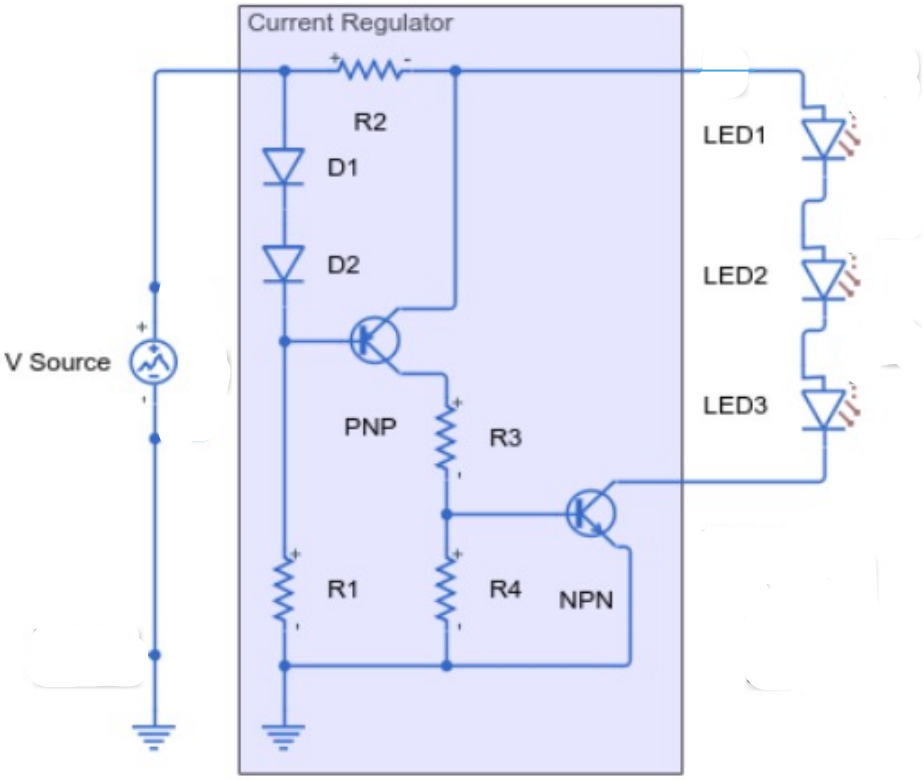
LED Circuits

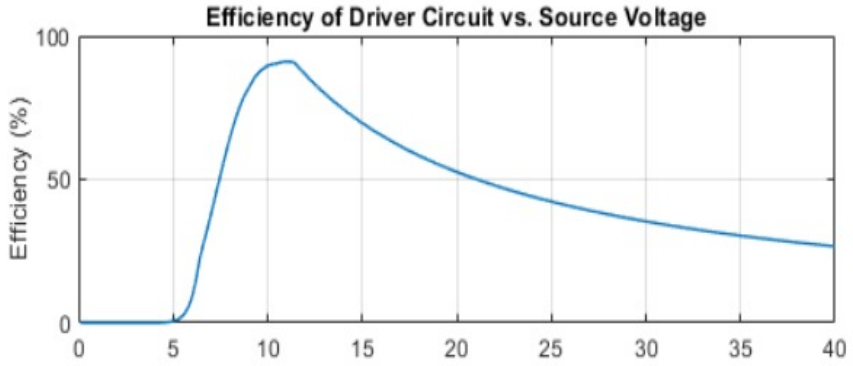
LED driver circuit in logical design. CMOS Logic provides 50 mA and 30 mA in the sink and source modes, respectively.

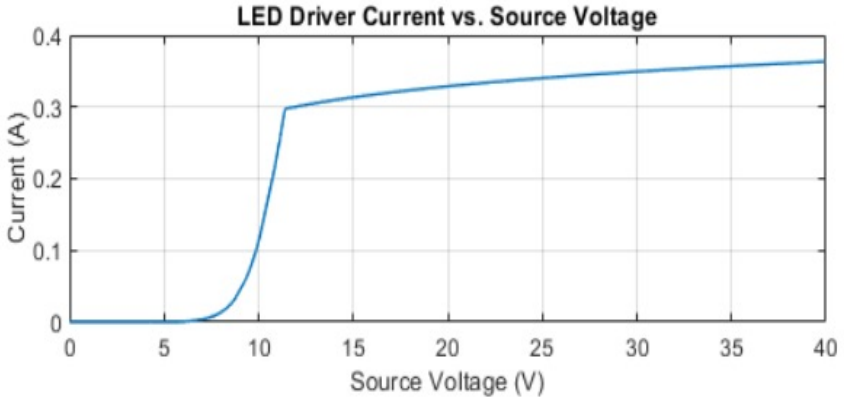


LED Circuits

LED light regulation circuits are used to keep light intensity constant against variations in input voltage.

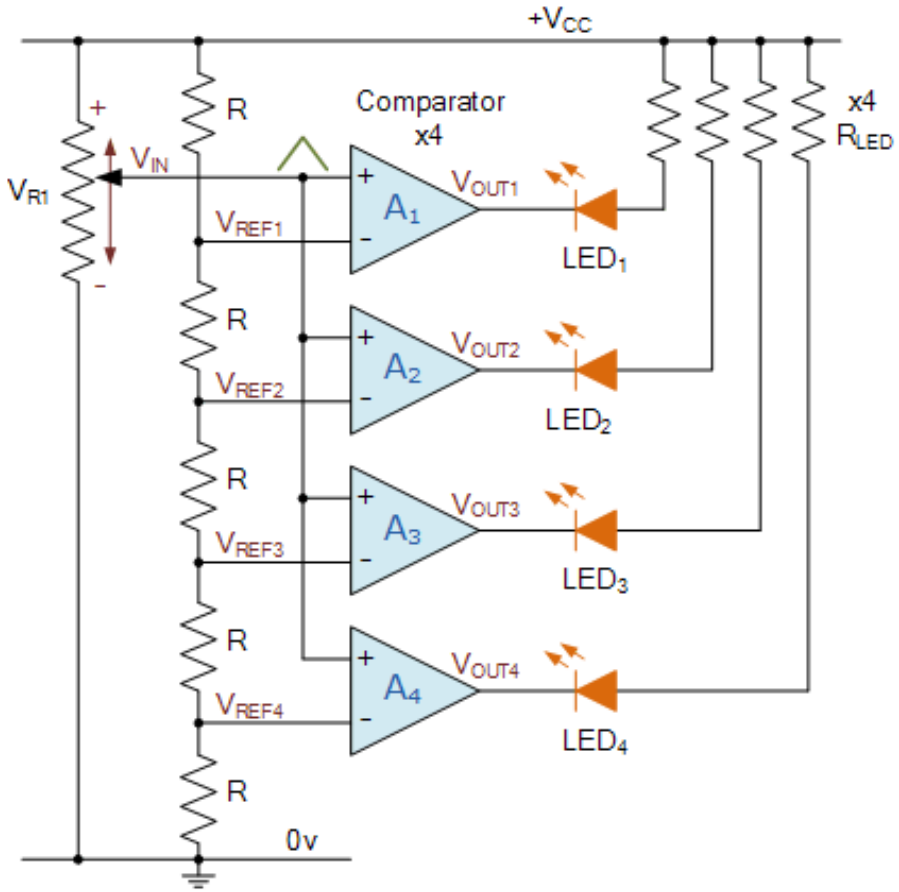






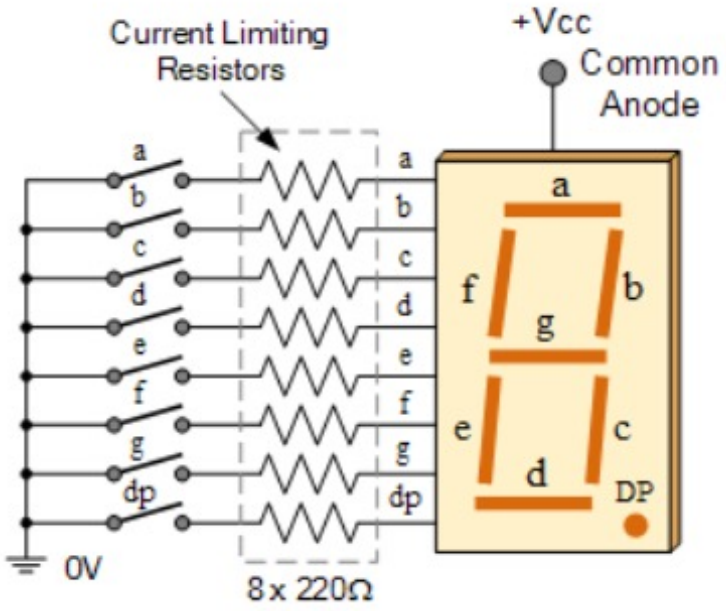
LED Circuits

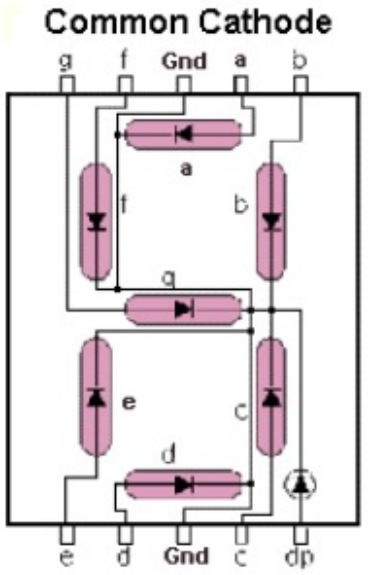
LED light brightness control circuit. The brightness of each LED is proportional to the input voltage level. Applications: Audio amplifiers circuits, Battery life check.

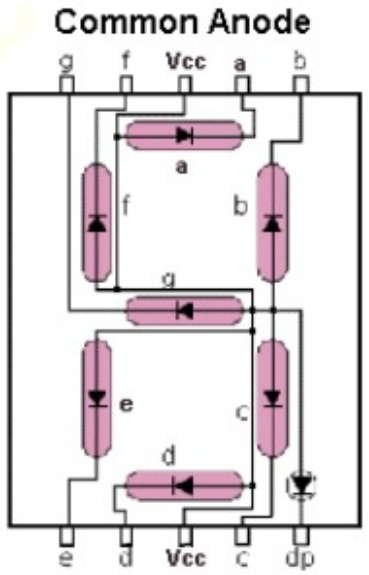


LED Circuits

The 7-segment Display

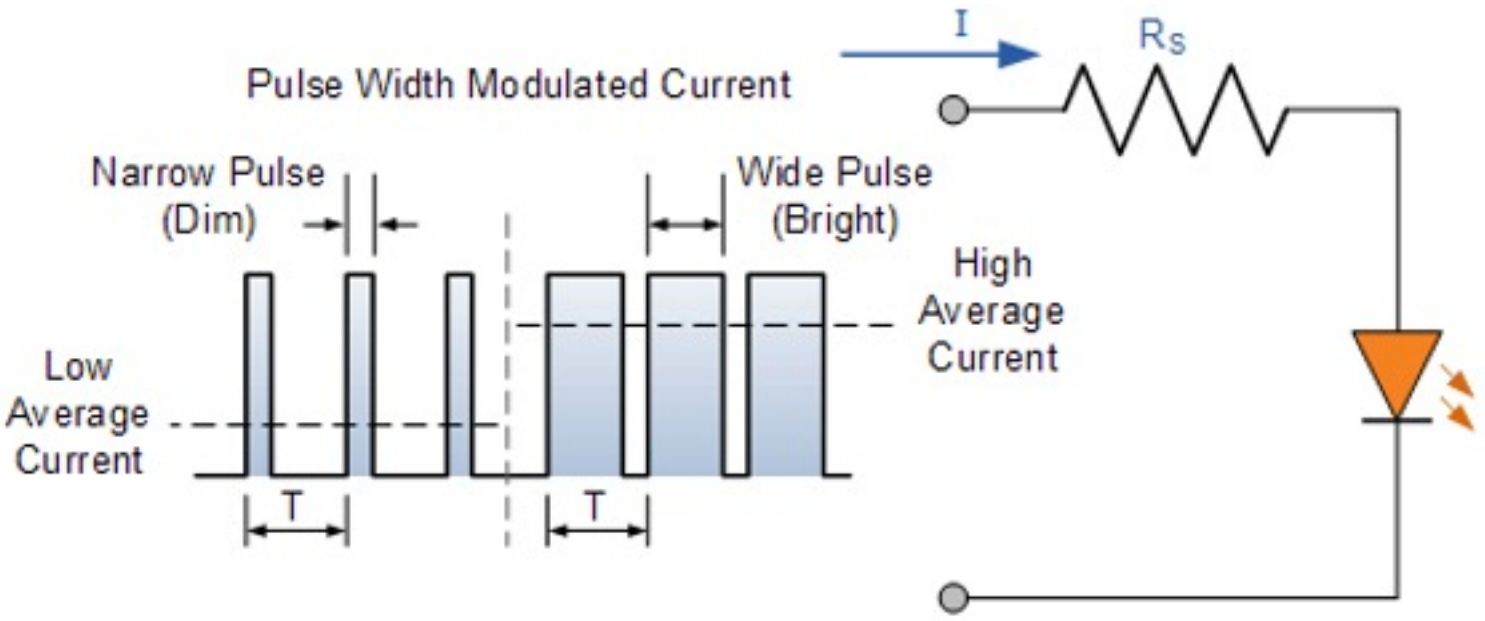






LED Circuits

LED refreshing circuit for power conservation. Pulses of frequency 100 kHz or higher produce light intensity that appears higher than continuous operation.



Q & A

Send me an email if you have any questions.